The effect of non-tariff barriers on unreported trade: Exploring a customs duty reform

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Abstract

Non-tariff barriers (NTBs) are a central yet understudied dimension of trade policy, particularly in their influence on customs duty compliance. Using detailed administrative import records and a Brazilian tariff reform, this paper shows that NTBs significantly reduce customs duty evasion by curbing value underreporting and virtually eliminating quantity misreporting. These effects vary substantially across firms, products, and trade partners: final consumption goods, non-manufacturing importers, and transactions with MERCOSUR countries exhibit weaker responsiveness to NTB enforcement. NTBs operate as an additional layer of regulatory enforcement, strengthening compliance incentives and shaping strategic trade reporting behavior.

keywords: non-tariff barriers, elasticity of customs duties evasion, tax enforcement.

JEL:F13, H26, F14, H25.

1 Introduction

Non-tariff barriers (NTBs) have emerged as a central issue in international trade policy, shaping the flow of goods across borders in ways that extend beyond traditional customs tariffs. These policy instruments—ranging from import licensing requirements to sanitary standards—can significantly influence both the volume and pricing of traded goods. Despite their increasing importance, the extent to which NTBs interact with customs duties to affect trade reporting discrepancies remains underexplored. This paper investigates the role of NTBs in mediating the relationship between customs duties and the unreported trade (gap)—the divergence between reported export and import values—an established measure of tariff evasion (Fisman and Wei 2004).

While a growing body of empirical research has examined the impact of NTBs on trade flows, linking these barriers to trade misreporting remains challenging due to limited data availability. This issue is particularly pronounced in developing economies, where comprehensive records on NTBs are scarce, and methodologies for identifying and quantifying their effects remain inconsistent. Researchers have relied on diverse approaches, including exporter surveys and trade policy reviews conducted by the World Trade Organization (WTO), to approximate the prevalence of NTBs (OECD 1997; OECD 2002a; OECD 2002b). However, these methods often fail to capture the full complexity of how NTBs shape trade reporting behavior.

We overcome these limitations by leveraging a unique dataset that provides detailed administrative records from Brazil at the transaction level. Our dataset encompasses all 8-digit product classifications, allowing us to construct a multidimensional panel of products with rich information on importers' characteristics, effective tariff rates, country of origin, border entry points, auditing coverage, and administrative treatments such as drawback regimes and tariff exemptions. We have precise data on NTB incidence, enabling us to examine their impact on trade misreporting with unprecedented granularity.

To compute our unreported trade statistic, we aggregate our data on imported goods, classified under the Mercosur Common Nomenclature (NCM), with all non-tariff barriers (NTBs) at the transaction level, allowing us to match them with exporters' reported values and quantities to the same 6-digit Harmonized System (HS) data in the UN-COMTRADE. The discrepant figures between our aggregated imported products and those reported by the exporters form the basis of our trade gap assessment, widely recognized in the literature as a measure of trade evasion.[1] Our empirical strategy relies on accurate taxes levied upon

[1]Therefore, we aggregated our import data originally at 8 to 6-digit harmonized system, producing shares of NTBs at that aggregated unit. Note also that not all discrepancies between export and import figures (trade gap or unreported trade) can be attributed to evasion. As highlighted by Mitikj and Kaushik (2024), the Discrepancy Index—based on asymmetries in trade reporting—offers valuable empirical insights into the nature and extent of such reporting differences. However, our focus is to estimate how these unreported trades respond to non-tariff barriers (NTBs) and align our findings with existing research that interprets these responses as the elasticity of customs duties evasion.

the arrival of goods at the border beyond the standard effective applied tariff. While the World Integrated Trade Solution (WITS) database provides detailed tariff information, it does not account for these additional import-related taxes in Brazil. Therefore, only by leveraging granular administrative data we can accurately assess the custom duties impact on unreported trade.[2]

We employ a many-way fixed-effects panel data model to isolate the effect of changes in customs duties on evasion behavior, exploiting within-group variation. Our identification of the elasticity of customs duties evasion (ECDE) hinges on the differential tariff exposure of imports following the introduction of two key taxes: PIS and COFINS. These taxes, implemented as part of the 2003 Tax Reform, were levied on virtually all imported goods.[3] Introduced with minimal lead time, they took effect less than six months after the reform's approval at rates of 1.65% and 7.6%, respectively. The swift implementation of these taxes left little room for strategic tax avoidance or adjustments to NTB regulations.[4]

We focus on the responses of the transactions based on their exposure to the NTBs after exogenous shifts in effective tariff rates. Given the comprehensive tax base of PIS and COFINS, we argue that the tax reform had a uniform impact across all products, regardless of their country of origin. Moreover, the imposition of NTBs is largely exogenous to changes in customs duties, reinforcing the robustness of our identification strategy. As NTBs can themselves be viewed as institutional features, we also control for other administrative variables (tariff agreement, anti-dumping, drawback, PIS/COFINS exemption, audits, exchange rate, trading companies, insurance and freight cost, number of imports and importers and the average company's years in activity and days until the import is cleared, states proportion of imports made through each Brazilian State) and state fixed effects. The administrative controls capture institutional characteristics specific to the product–country level, while the state fixed effects account for geographic and port-of-entry differences that may reflect distinct institutional practices or special treatments.

We introduce an intuitive framework that reinforces NTBs as a modulator of enforcement quality. As their imposition captures an increase in compliance cost to the importers, we expect a negative influence on evasion responses. Then, we discuss our findings without accounting for NTBs to facilitate a comparison of our estimates with the previous literature on ECDE. Our results of the unreported trade responses to tariff changes suggest an elasticity of 1.98 for values and 0.97 for quantities — larger and more precisely estimated

[2]Note that this is a typical cascade tax system, common in developing economies in the period (Keen 2014). See also Kume, Piani, and Miranda (2011) and Sousa, Tannuri-Pianto, and Santos (2008) that explore the COMTRADE data for Brazil. Recently, Bussy (2023) investigates the impact of export-import misreporting on corporate tax evasion.

[3]Constitutional Amendment No. 42, of December 19, 2003. Available at planalto.gov.br. The taxes began to be collected on May 1, 2004. See (Werneck 2006).

[4]Only a limited set of products qualified for zero tax rates (e.g., hydraulic motors, cylinders, parts of harvesting machines). In contrast, others were fully exempt (e.g., books, periodicals, and paper for printing newspapers).

than models relying solely on publicly available, effectively applied tariff rates.[5] When we include the interaction between non-tariff barriers and tariff rates, the ECDE decreases to 0.76 for values and -0.09 for quantities, reinforcing the role that NTBs work as a pre-auditing policy effective in reducing unreported trade activities. Our findings also emphasize that product reclassification—shifting goods into lower tariff categories—plays a key role in evasion behavior.[6]

Last, our heterogeneity results suggest that manufacturing firms mitigate unreported trade responses to tariff reforms. This aligns with our evidence on inputs and capital goods; the primary components of manufacturing imports exhibit lower unreported responses than consumption goods, which are typically linked to the retail sector. Moreover, Mercosur imports consistently show stronger estimated unreported responses to NTBs and tariffs.

We contribute to the literature in three main aspects. First, we add to the literature on non-tariff measures - NTM, WTO (2012) - usually associated as a substitute for tariffs in the welfare gains from trade liberalization (Grübler and Reiter 2021; Egger et al. 2015; Ferrantino 2006; Kinzius, Sandkamp, and Yalcin 2019; Bao and Qiu 2010; Winters 1987; Dick 1994)[7].

Our investigation focuses on a primary risk factor that contributes to customs duty evasion. Specifically, we focus on non-tariff barriers that apply to approximately 25% of Brazilian imports. The NTBs include sanitary, phytosanitary (SPS), Technical Barriers to Trade (TBT), quantitative restrictions (i.e., quotas), and other measures. In Brazil, these NTBs are encompassed by an administrative procedure known as non-automatic import licensing (henceforth, import license). The import license requires the importer to submit an electronic form before shipping the goods, with government authorization for imports contingent upon verification of compliance with legal and administrative regulations. Although various administrative treatments for imports are associated with tariffs, such as tariff agreements, drawback regimes, and exemptions, it is important to note that import licenses are categorized as a non-tariff barrier and apply to a consistent

[5]Using only WITS (publicly available) data to calculate tariff rates and tax amounts at the 6-digit HS level for Brazil, instead of our administrative data, results in a lower effect on value (1.12) and quantity (-0.11), the latter not significant. See also Choi and Yu (1987) that explores the impact of tariffs on terms of trade and on the import demand.

[6]Our estimation of the elasticity of customs duties evasion aligns with existing evidence. Mishra, Subramanian, and Topalova (2008) report 0.1 for India; Javorcik and Narciso (2008) find 0.4 for homogeneous goods and 1.7 for differentiated products in German and Eastern European data; and Javorcik and Narciso (2017) show elasticities of 0.65 (values) before WTO accession and 1.1 (quantities) after. Fisman and Wei (2004) document about 3 for China–Hong Kong trade. Our results are consistent with these studies and highlight the moderating role of NTBs.

[7]Import licensing and pre-shipment inspections can have similar interpretations for the impact on ECDE (Yang 2008; Anson, Cadot, and Olarreaga 2006). *Ad Valorem* Equivalents (AVE) are estimated for NTM to derive trade elasticities, which are the key statistics to calculate gains from trade. The role of non-tariff measures in affecting trade flows and fraud is much less explored. See Kee and Nicita (2016) as an exception to this literature. While the presence of trading companies is identified as one of the relevant sources of heterogeneity, it does not exhibit a statistically significant effect in our linear model specification. Detailed results for trading companies can be found in the previous version of our paper, and these results are available upon request.

share of imports throughout our study period.[8]

Second, we contribute to the literature that estimates ECDE (Fisman and Wei 2004; Javorcik and Narciso 2008; Mishra, Subramanian, and Topalova 2008; Bouet and Roy 2012), as we combine the impact of both tariff and non-tariff policies on import reporting. While the literature on customs duties evasion has explored either the level of tariffs (Fisman and Wei 2004), the role of differentiated products (Javorcik and Narciso 2008; Mishra, Subramanian, and Topalova 2008) as a relevant risk factor, little has been investigated about other widespread importing policies that may have a modulating effect on the elasticity of evasion. This study aims to enhance our comprehension of tariff evasion responses to the combination of tariffs and non-tariff policies.

Third, analyzing Brazil offers insights applicable to other (large) emerging economies.[9] As Latin America's largest economy, Brazil's import volume reached US$292.3 billion in 2022, with trade comprising 19.3% of its GDP. Its diverse import portfolio and ties with 219 countries make it an ideal setting for studying import duty evasion, with findings that can inform both domestic and international policy.

The remainder of this paper is organized as follows. The next section provides a conceptual framework for our analysis. Section 3 presents details of the datasets used in this article with descriptive statistics and our empirical specification. Section 4 brings in our results, including heterogeneity analysis. Finally, we conclude in section 5.

2 Conceptual Framework

This section presents a simple model that captures the impact of enforcement and tax-related knowledge on tariff evasion elasticity. Our model draws from Mishra, Subramanian, and Topalova (2008), Slemrod (2001), and Yang (2008) and assumes that importers are rational agents who optimize behavior as in Allingham and Sandmo (1972). This framework guides us to interpret the effect of NTB on evasion, considering that it acts as an enforcement enhancement. Misreporting imports entails a risk, modeled in reduced form as a cost function that captures the expected cost of evasion. Assume a cost of evasion as follows:

$$\tilde{C} = \tilde{C}(\gamma, E)$$

The "smuggling" fraction, represented by the variable γ, is a value between 0 and 1, and E is a measure of enforcement quality, empirically measured by the exposure to the NTB at the origin-country-product-year level. $\tilde{C}(\gamma, E)$ captures the part of the cost of evasion,

[8] We also employ a machine learning model using causal forests to validate the selection of non-tariff barriers as a primary source of heterogeneity in the elasticity of customs duties evasion, see Appendix 4.

[9] See Beverelli et al. (2023) for the role of a country-specific institution as a determinant of international trade.

which is independent of tariffs but related to the probability of detection.[10] Therefore, the quality of enforcement, E, captures any additional enforcement capacity available to the tax authority that is not directly related to tariffs. For instance, E can be associated with product characteristics that facilitate valuation (Rauch 1999; Javorcik and Narciso 2008) or with the information about the product, producer, and terms of trade provided before shipment through processes such as import licenses or pre-shipment inspections — non-tariff measures. The following assumptions are made about this cost[11]: i) $\tilde{C}(\gamma, E) \geq 0$, cost of evasion is non-negative; ii) $\tilde{C}_1(\gamma, E) > 0$, the marginal cost of evasion is positive; iii) $\tilde{C}_2(\gamma, E) > 0$, the cost is increasing in enforcement; iv) $\tilde{C}_{11}(\gamma, E) > 0$, the convexity of costs to evade and; v) $\tilde{C}_{12}(\gamma, E) > 0$, the marginal cost of evasion is increasing in enforcement[12].

With this specification, the evasion cost does not depend on the tariff rate, as in Yitzhaki (1974).[13] This is consistent with Brazilian legislation that establishes that if an importer is caught illegally undervaluing an imported good (say, forging the invoice or smuggling the good), the penalty is based only on the amount by which the good is undervalued. However, when evasion occurs through misclassification, the penalty is proportional to the amount of the evaded tariff.

We extend this model to consider that the probability of detection and punishment is not related to tariffs, and to support this assumption, our Figure 1 shows no sign of a clear structural relationship between the two variables with an average audit rate of 24%, i.e., the probability of being audited, and therefore caught evading, is unrelated to the tariff rate.[14]

Therefore, we augment our cost function to linearly add a term incorporating a representative firm that imports a constant amount M, a fixed probability of punishment, $\bar{p}$, the tariff rate represented by T, and a penalty parameter θ that means that the importer's cost when caught evading is proportional to the duty evaded.[15] Our total cost of evasion

[10]This smuggling can occur in different forms: (i) Underpricing, (ii) misclassification, or even (iii) concealing quantities.

[11]Such assumptions in cost of evasion are common in the literature, (Cowell 1990; Kaplow 1990; Mayshar 1990; Hines and Rice 1994; Slemrod 2001; Yang 2008; Chetty 2009; Mishra, Subramanian, and Topalova 2008)

[12]We use the following notation to represent partial derivatives, $\partial\tilde{C}/\partial\gamma = \tilde{C}_1$, since γ is the first argument to the cost function. Second-order derivatives will have the same notation with an additional index, $\tilde{C}_{12} = \partial^2\tilde{C}/\partial\gamma\partial E$.

[13]We emphasize that our model provides the foundation for the empirical analysis by explicitly accounting for a broad set of heterogeneous factors, including product characteristics, country of origin, and time-specific effects. The empirical strategy incorporates an extensive set of fixed effects to robustly control for these dimensions. Furthermore, the heterogeneous results and robustness checks—based on product- and firm-level characteristics, as well as trade bloc origin—reinforce the consistency and reliability of our main findings without altering their qualitative interpretation.

[14]We also estimate a linear regression of audits on tariffs at the product level. We consistently find small, statistically insignificant coefficients, indicating no direct association between tariff rates and audit likelihood. Table 2.1 in Appendix 2 presents the results.

[15]This assumption is made for simplicity, as in Mishra, Subramanian, and Topalova (2008). In their work, they show that as long as the absolute value of the elasticity of imports concerning *tariffs* is less

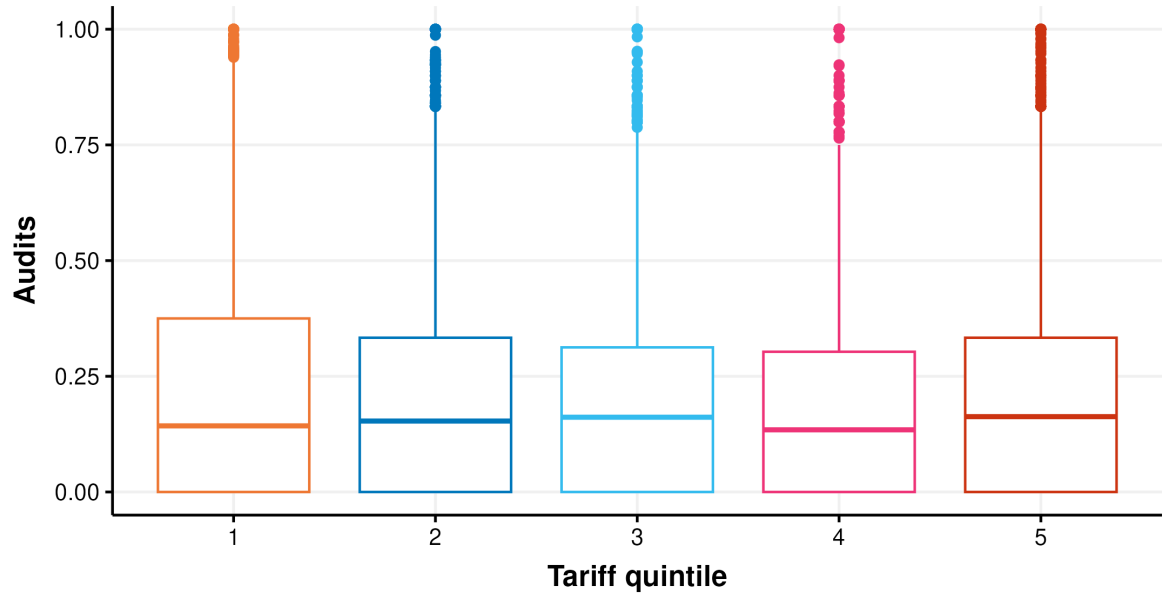


Figure 1: Proportion of imports audited by tariff quintile and year. Box-plots indicating 25th, 50th and 75th quantiles.

has the following specification:

$$C(\gamma, E, T) = \tilde{C}(\gamma, E) + \bar{p}\theta\gamma MT.$$

This typical importing firm maximizes the net benefit of evasion, which is the difference between the duties avoided γMT and the cost of evasion, given by $B = \gamma MT - C(\gamma, E, T)$. Therefore, firms must choose the optimal "unreported" fraction γ^* to evade/avoid customs duties. The first-order condition, which implicitly defines the level of optimal unreported trade as a function of the tariff rate T and enforcement E, is given by:

$$MT(1 - \bar{p}\theta) = \tilde{C}_1(\gamma^*, E). \tag{1}$$

This equation represents the demand side of evasion and has a simple interpretation: evasion's marginal cost must equal its marginal benefit. Differentiating equation (1) with respect to T leads us to the slope of this demand function for the tariff rate:

$$\frac{\partial \gamma^*}{\partial T} = \frac{M(1 - \bar{p}\theta)}{\tilde{C}_{11}(\gamma, E)} > 0. \tag{2}$$

Our assumption of convex costs in evasion implies that $\partial\gamma^*/\partial T > 0$, the semi-elasticity of duties evasion concerning the tariff rate, is positive. Therefore, as long as $\bar{p}\theta < 1$, which is

than one, the results are unchanged. In Appendix 1.2, we show that, with minor assumptions, if imports respond to enforcement policy, our results also hold.

a plausible assumption based on our data, the elasticity of evasion is damped by a factor of $1 - \bar{p}\theta$ and, is positively related to the tariff exposure.[16]

Next, by differentiating the equation (1) with respect to the enforcement variable, we can see how the optimal evasion level is affected by enforcement quality. The assumptions (i) of convex evasion costs and (ii) increasing marginal costs with enforcement quality (i.e., $\tilde{C}_{12}(\gamma, E) > 0$) ensure that $\partial\gamma^*/\partial E < 0$.

$$\frac{\partial\gamma^*}{\partial E} = \frac{-\tilde{C}_{12}(\gamma, E)}{\tilde{C}_{11}(\gamma, E)} < 0. \tag{3}$$

Therefore, the evasion decision in our model increases with the tariff rate and decreases with the quality of enforcement. This framework can also be used to understand how enforcement modifies the elasticity of evasion in relation to the tariff rate. Taking the derivative of evasion elasticity in relation to the enforcement proxy to calculate how the evasion response to tariff rate changes with enforcement leads us to:

$$\frac{\partial^2\gamma^*}{\partial T\partial E} = -\frac{M(1 - \bar{p}\theta)}{\tilde{C}_{11}^2(\gamma, E)}\frac{\partial\tilde{C}_{11}(\gamma, E)}{\partial E}. \tag{4}$$

Using the standard assumption that the convexity of the cost of smuggling increases with the enforcement level, i.e., $\partial\tilde{C}_{11}(\gamma, E)/\partial E > 0$, implies that $\partial^2\gamma^*/\partial T\partial E < 0$ means that the elasticity of evasion is reduced by better government monitoring. This result, estimated in the main findings below, reinterprets unreported trade responses to tariff rate changes as influenced by enforcement quality variations captured through our NTBs.

3 Empirical Implementation

3.1 Data

We consider two primary sources of data. The first brings all reported imports (and exports) by the Brazilian firms (from all their partners) for 2001 to 2004, aggregate at HS 6-digit, from the WITS/COMTRADE databases. We label this dataset the *"Trade sample"*. Second, we use a confidential administrative dataset that comprises all Brazilian imports at the transaction level, our *"Administrative sample"*. This data is sourced from the Ministry of Economy and extracted from the SISCOMEX information system, which records all import and export processes in Brazil using 8-digit product codes based on the NCM standard.[17] The import values are FOB, thus directly comparable to export data from the trade sample. Using the detailed information of the administrative sample (US dollar

[16] Using our administrative dataset, the probability of audit is close to 23.45%, while the penalty term in Brazil is typically 75%. This leads to a typical $\bar{p}\theta \approx 0.18$.

[17] The Integrated Foreign Trade System - SISCOMEX is an administrative information system that integrates the activities of registration, monitoring, and control of foreign trade operations in Brazil.

import values, quantities, importer firm, final versus intermediate products, purchaser, exporter, most favored nation tariff level, effectively applied tariff rate - including all tariffs and taxes at the border, exemption status, preferential trade agreement, import license, transaction auditing), we can compute the effective average tariff rate for 6-digit HS codes by country-product-year.[18] More importantly, import duty (II) is not the only tariff that imports are subject to in Brazil. Our administrative data is the only source that allows us to compute all other border duties, such as the Industrialized Products Tax (IPI), the Social Integration Program (PIS), and the Contribution for the Financing of Social Security (COFINS), on imports. When using our trade sample exclusively to compute tariffs, we use the Most Favored Nation (MFN) Tariffs, as they refer to the standard tariff rates that Brazil applies to imports from other World Trade Organization (WTO) member countries, unless a preferential trade agreement (such as a free trade agreement or customs union) implies lower rates. These tariffs are based on the MFN principle, which ensures that WTO members do not discriminate among their trading partners by offering preferential tariffs to some while imposing higher tariffs on others, except in cases where specific trade agreements permit such preferences. **Analysis Period.** Our datasets, *administrative and trade*, cover all Brazilian imports during 2001-2004. We consider the restricted sample of 2001-2002 in our placebo-like exercises as a parallel-trend assumption test for any anticipation effect of the reform. As the PIS and COFINS on imports were established on May 1st, 2004, exogenously by the Federal Act 10865/2004, we focus on the short period around this tax reform (2003-2004). That is the case because many subsequent changes in the effective import taxes after 2005 reduced our conviction on the exogeneity of this shock for periods beyond 2004.[19] Therefore, this paper investigates short-term evasion responses to this tax reform.

Evasion measures: Unreported Trade. We build two measures of value evasion and two of quantity evasion of customs duties (Fisman and Wei 2004), labeling them as trade-gaps.[20] For each of those measures, we combine our two datasets, aggregating our administrative sample at 6-digit codes to match the export data from the UN COMTRADE data as in Mishra, Subramanian, and Topalova (2008).[21]

[18]We identify the nature of importing firms, whether trading companies or not. Trading companies are (i) registered purchasers in Brazilian customs regulations, and (ii) they have imported more than twenty different products using the 8-digit classification code, and each product has been imported more than twenty times. We also consider two alternative definitions to assess robustness to our benchmark choice, high and low pervasiveness (25/35 and 10/15, respectively), and our results are not qualitatively affected by those definitions.

[19]Federal Act 10865/2004 consolidated link can be accessed at http://www.planalto.gov.br/ccivil_03/_Ato2004-2006/2004/Lei/L10.865.htm.

[20]We drop approximately 1.94% and 2.99% of value and quantity observations, respectively, due to non-matched exports. This trade gap measure introduced in Fisman and Wei (2004) has since been the standard proxy variable for evasion in border duties literature, (Mishra, Subramanian, and Topalova 2008; Javorcik and Narciso 2008; Bouet and Roy 2012; Javorcik and Narciso 2017).

[21]Even though we have data on imports at the transaction level, we need to consolidate our administrative dataset by country, product, and year to align it with COMTRADE export data. We conduct the regression analysis based on years, considering 2004 as the post-shock year.

$$Ev_{cpt} = \log(X_{cpt}) - \log(M_{cpt}) \tag{5}$$

Where Ev_{cpt} is our value (quantity) gap, and X_{cpt} are the exports to Brazil reported by partner country c of the product p at year t and M_{cpt} are the import values as declared in Brazil upon goods arrival carefully measuring trade in the same units. This evasion measure restricts transactions for which there are matched non-zero exports and imports. There is a corresponding import for every export transaction with a value greater than zero.[22] The second evasion measure assumes that the value (or quantity) imported is assumed to be zero for any reported export that does not have a matching import and captures the extreme value (quantity) evasion. This extreme assumption needs a slight modification in the equation (5) as $Evx_{cpt} = \log(1 + X_{cpt}) - \log(1 + M_{cpt})$, and our sample must include those observations for which exports are recorded with a value greater than zero, but no counterpart import transaction is recorded.[23]

Non-tariff Barriers. An NTB is a specific case of Non-tariff measures (NTM) where the effect is known to reduce trade, thus imposing a barrier on free trade. These measures can be broadly divided into two groups. Technical measures include regulations, standardization, testing, and certification procedures. They are primarily sanitary, phytosanitary (SPS), and Technical Barriers to Trade (TBT) measures. Nontechnical procedures include quantitative restrictions (i.e. quotas), price measures, and forced logistics or distribution channels.[24] In Brazil, these non-tariff barriers are generally classified as import licensing as this encompasses SPS, TBTs, quotas, and all barriers where a regulatory agency reviews the import process. We cannot distinguish between them in our analysis, although we acknowledge that all of them pose an impediment to free trade. Therefore, we concentrate on these NTBs, referring to them interchangeably as import license, which affects approximately 25% of Brazilian imports, as outlined in Table 1.

The import license is an electronic document through which the government authorizes the import upon verification of compliance with legal and administrative rules and is required when the import is subject to the approval of one or more consenting bodies (Verification Authorities). It depends on the product, country of origin, condition of the product (e.g. used or brand new), and the corresponding taxation regime. This process may take up to 60 days to complete, and the importer is not authorized to initiate the shipment process without such a license. The importer is subject to penalties up to 30% of the import CIF (Costs-Insurance and Freight) value in case of not complying with the rule.[25]

[22] Note that whenever the Administrative sample is used, the values (or quantities) imported, M_{cpt}, come from this dataset. In contrast, export figures always come from the Trade sample.

[23] We also run mid-point and first difference trade gap regressions. Results are available upon request.

[24] Details on NTM can be found at https://www.oecd.org/trade/topics/non-tariff-measures/. The anti-dumping measure is one of the non-tariff barriers in our administrative dataset but is not used as an NTB, representing only 0.003%.

[25] The free-on-board value is added by the insurance and freight costs.

Henceforth, the imposition of a non-tariff barrier could plausibly increase the quality of enforcement facilitated by providing detailed information to customs authorities. Importers may perceive the entire process of obtaining an import license, contingent upon submitting comprehensive information to the government, as an elevated level of enforcement.

Table 1: Descriptive Statistics

	2003			2004		
Variable	Mean	SD	Obs.	Mean	SD	Obs.
Value Gap	0.1019	2.0715	29,619	0.1069	2.1065	31,311
Extreme Value Gap	0.0824	1.7975	31,399	0.0712	1.8384	33,323
Quantity Gap	0.0209	2.5547	29,278	-0.0315	2.5602	30,970
Extreme Quantity Gap	0.0206	2.9745	31,266	-0.0365	3.0103	33,195
Tariff Exposure	0.1671	0.1162	55,476	0.2278	0.1230	57,809
Import License	0.2522	0.3790	55,476	0.2493	0.3785	57,814
Tariff Similar	0.1176	0.1065	55,476	0.1686	0.1109	57,804
Anti-dumping	0.0003	0.0142	55,476	0.0002	0.0127	57,814
Audits	0.2422	0.2883	55,476	0.2272	0.2830	57,814
Drawback	0.0462	0.1638	55,476	0.0497	0.1671	57,814
Tariff Agreement	0.0512	0.2077	55,476	0.0503	0.2058	57,814
Ex-II	0.0046	0.0523	55,476	0.0057	0.0578	57,814
Ex-IPI	0.0038	0.0487	55,476	0.0031	0.0428	57,814
PIS/COFINS exemption	0.0167	0.1274	55,473	0.0167	0.1276	57,808

Notes: Trade gaps are given in log-dollars for value or log-quantity. Tariff exposure is the total tariff exposure, comprising the II, IPI, PIS and COFINS taxes. All other variables from Import License to PIS/COFINS exemption represent the proportion of imports subject to that administrative treatment.

Trade-Gap statistics. Table 1 shows our measures of the trade gap using administrative data. We report that trade gaps will generally be positive or near zero, as importers often want to obscure their transactions. Our administrative tax data reinforces that evasion indicators exhibit the expected positive sign, except for negative values for the quantity gaps in 2004. We split the table by year to better comprehend the tariff shock.[26]

The variable non-tariff barrier examined in this paper is captured by the broad definition of import licenses.[27] Other administrative treatments include the drawback regime, Ex-II, Ex-IPI, exemption of PIS and COFINS on imports, and tariff agreement (e.g. MERCOSUR), all used as control variables.[28] Audits represent the proportion of imports that were

[26]See our previous version of this paper https://econpapers.repec.org/paper/fgveesptd/560.htm, Table 9, for the comparison of trad-gap statistics using our administrative data versus the Trade Data.

[27]We find that approximately 0.03% of imports were subject to anti-dumping measures, the only additional NTB in our dataset.

[28]The drawback is a special customs regime that allows for the suspension or elimination of taxes levied on the acquisition of inputs used in the industrialization of exported products. Ex-II and Ex-IPI refer

submitted for physical inspection. Finally, Table 1 reveals the evolution of the Tariff Exposure and Tariff of Similar goods between 2003 and 2004 from 16% to 22% and 11% to 16%, respectively.[29] This contrasts with the relative stability of the import license measure around 25%. Note also that the average audits across the two years have slightly decreased.

Table 2: Summary statistics separated by NTB status.

	Preshock (16 months)			Postshock (8 months)			
Variable	Mean	SD	Obs.	Mean	SD	Obs.	Norm. Diff.
No NTB							
Tariff Exposure	0.1802	0.1234	4,231,864	0.2773	0.1452	2,568,464	0.7204
Anti-dumping	0.0000	0.0059	4,231,864	0.0000	0.0054	2,568,469	-0.0008
Audits	0.2046	0.4034	4,231,864	0.1675	0.3734	2,568,469	-0.0954
Tariff Agreement	0.0277	0.1641	4,231,864	0.0278	0.1645	2,568,469	0.0008
Ex-II	0.0012	0.0342	4,231,864	0.0031	0.0558	2,568,469	0.0422
Ex-IPI	0.0082	0.0902	4,231,864	0.0035	0.0588	2,568,469	-0.0622
PIS/COFINS exempt	0.0189	0.1362	4,231,857	0.0198	0.1392	2,568,464	0.0062
NTB							
Tariff Exposure	0.0355	0.0942	1,635,698	0.1240	0.1043	1,001,117	0.8901
Anti-dumping	0.0003	0.0185	1,635,698	0.0002	0.0143	1,001,120	-0.0084
Audits	0.1988	0.3991	1,635,698	0.1691	0.3748	1,001,120	-0.0768
Tariff Agreement	0.0623	0.2418	1,635,698	0.0588	0.2352	1,001,120	-0.0151
Ex-II	0.0079	0.0886	1,635,698	0.0057	0.0751	1,001,120	-0.0272
Ex-IPI	0.0064	0.0796	1,635,698	0.0044	0.0663	1,001,120	-0.0268
PIS/COFINS exempt	0.0573	0.2323	1,635,680	0.0552	0.2284	1,001,117	-0.0088

Notes: Tariff exposure is the total tariff exposure, comprising the II, IPI, PIS and COFINS taxes. All other variables from Anti-Dumping to PIS/COFINS exemption represent the proportion of imports subject to that administrative treatment. The preshock period ranges from 2003-01-01 to 2004-04-30 and postshock covers the dates from 2004-05-01 to 2004-12-31.

Our Table 2 provides summary statistics comparing imports subject to NTBs during pre and postshock periods and those exempted from such a procedure. It reveals the following key findings: (i) Products that are subject to NTBs tend to have lower tariff rates, aligning with the observations previously documented in Kee and Nicita (2016); (ii) Audits have shown a similar decline rate between two distinct groups: those affected by NTBs and those not affected by NTBs, spanning

to exception tax rate cases for imported and industrialized products, respectively. PIS and COFINS on import exemption status are based on the authors' interpretation of the legislation to set specific HS codes as exempt. For the pre-shock period, the exemption status is counterfactual. Imports that would have been exempt had PIS and COFINS been implemented at the time of import registration.

[29]The variable *Similar* controls for potential misclassification and tariff evasion strategies by importers. Firms seeking to underreport imports after tariff changes may deliberately misreport the classification of imported goods, declaring them under product categories with lower tariff rates to reduce their tax burden and claim the lowest tariff among similar goods.

the pre-shock to post-shock periods. This suggests that the impact of audits on these two groups has been comparable over time.

Additionally, the proportion of imports subject to a tariff agreement is slightly higher in the NTB group compared to the No NTB group. This discrepancy is primarily driven by agricultural products, which are typically subject to NTBs. In the case of Brazil, the country is a significant importer from its MERCOSUR partners[30].

The exogeneity of PIS/COFINS shock. The Constitutional Amendment 42/2003 and Federal Act 10833/2003 brought about a significant transformation in the taxation scenario, particularly concerning PIS and COFINS, extending the scope of these taxes to include imports, previously exempt.

The new rates for noncumulative sectors were set at 1.65% and 7.6%, respectively. In Brazil, customs duties must be fully settled at nominal rates (e.g., 1.65% for PIS and 7.6% for COFINS) before goods can be released from the customs area. Importers are obliged to remit the Import Duty (II), the Industrialized Products Tax (IPI), PIS, COFINS, and the State Value Added Tax (ICMS) in their entirety before the transportation of goods to their respective facilities.[31]

Although the PIS and COFINS rates levied upon imports were generally the same for all products, the differential exposure to this shock comes from previous rates of II and IPI. PIS and COFINS are "cascading" taxes, as their base includes other import duties. The incidence chain of border duties is as follows: import tax II is applied to the CIF value; then, the tax on industrialized products, IPI, is levied upon the sum of the CIF value and the amount paid for II; finally, both PIS and COFINS are levied upon the whole sum of the CIF value, II, and IPI. Therefore, the composite tariff rate after the introduction of PIS and COFINS is given by $T_p = (1+II_p)\times(1+IPI_p)\times(1+PIS+COFINS)-1$, where we make it clear that II and IPI rates are dependent on the product p being imported. Considering that in 2003, PIS and COFINS were nonexistent, we get the expression $\Delta T_p = T_{p,2004} - T_{p,2003} = (PIS + COFINS) \times (1 + II_p) \times (1 + IPI_p)$ for the tariff variation induced by the introduction of the new border duties.[32]

Last, our Figure 2 shows the scatter plots for the Most Favored Nation (MFN) and the effectively applied tariff rates (EAR) tariff burden, with their figures plotted on a reference line of slope one. We can observe that most of the products lie above this reference line, demonstrating the increase in burden due to the PIS and COFINS shock.[33] Although there is a burden reduction for some products, for example, in the EAR (MFN) top panel, 3.00% (2.38%) of products had tariff exposure reduction in the post-shock period. The correlation of tariff exposure is positive and close to one, reinforcing the argument of an exogenous shock. If compensation in II and IPI rates had fully occurred, we would observe a flattening predictive curve where the 2004 exposure would have little correlation to the previous year's tariff rate. However, a perfect and positive correlation

[30] As an example, in 2003, Brazil accounted for 85.9% of Argentina's wheat exports according to The Observatory of Economic Complexity (https://oec.world/en).

[31] For a product subject to the following typical tax rates: 5%, 15%, 1.65%, 7.6%, and 18%, respectively, the cumulative tax burden would exceed 55% of the import value.

[32] We calculate the average II and IPI rates for each product and categorize these products into quartiles based on these rates. We calculate the average exposure in the post-shock period for products falling within these quartiles to demonstrate that there is no change in average rates for all imports or NTB imports. See Table 2.5 in the Appendix 2.

[33] MFN tariffs come from WTO, and EAR comes from our calculation of applied tariffs.

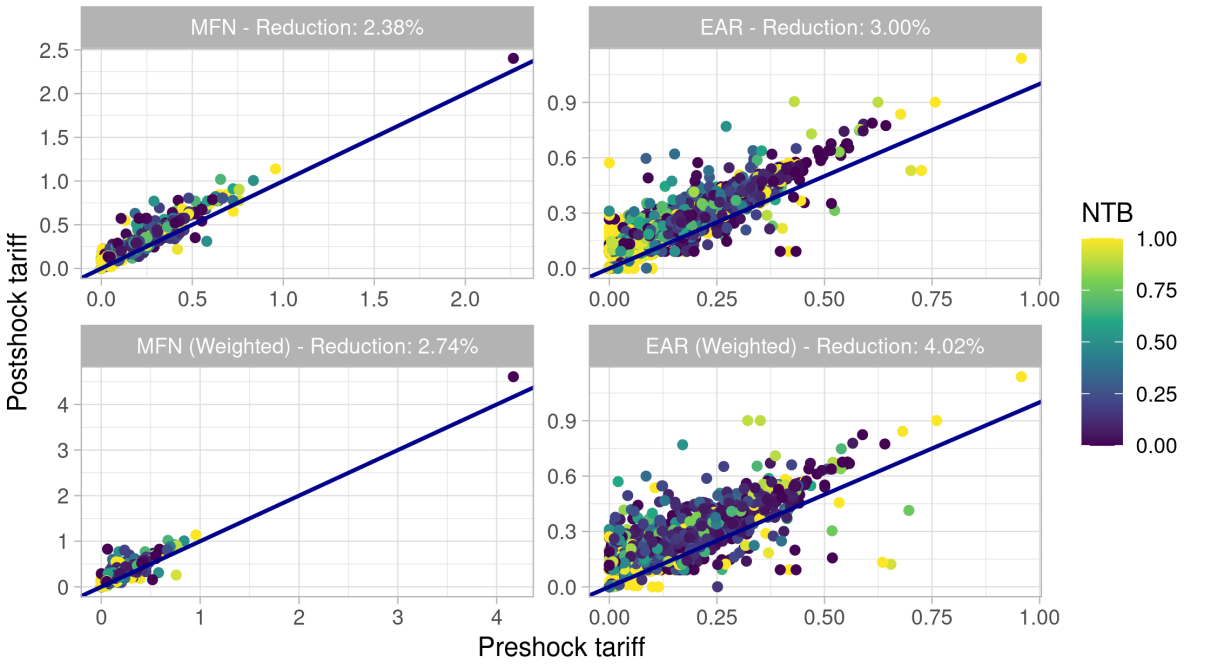


Figure 2: Correlation of pre and post-shock tariff burden for non-weighted (Panel A) and weighted (Panel B) figures. Each dot represents an HS 6-digit product. Most favored nation (MFN - public data) and effectively applied tariff rates (EAR -administrative data). Products above the reference line have experienced a post-shock increase in total tariff exposure. The top line shows the fraction of products that experienced decreased tariff burden for each sample. Previously, high-taxed products tended to keep their status in the post-shock period. The color scheme denotes the pre-shock proportion of NTB on the number of imports. The top row presents tariff exposure computed from simple averages, and weighted averages are used in the bottom row. Each panel discriminates the share of observations that reduced exposure pre- versus post-shock.

indicates no compensation; the most taxed goods in 2003 are still at the top of the rankings after the shock, with a higher total tariff exposure.

The findings displayed in Figure 2 reveal that the pre-shock proportion of non-tariff barrier (NTB) imports, depicted by lighter colors (i.e., green-yellow), does not show any noticeable pattern. This observation implies that businesses importing products subject to NTBs have not significantly changed their product mix in the post-shock period.

3.2 Econometric specification

Unreported imports can occur through the manipulation of the unit value of imported goods, smuggling larger quantities, or misclassifying products within the NCM (Mercosur Common Nomenclature). We employ a machine learning model based on the causal forests algorithm (Athey, Tibshirani, and Wager 2019; Wager and Athey 2018) to explore the potential factors contributing to border duty evasion. Our analysis reveals that, in three out of four evasion measures, the most influential feature for explaining evasion heterogeneity is the similarity in product classification.[34]

To incorporate product misclassification into our econometric approach, we adopt the methodology outlined by Fisman and Wei (2004). For each observation at the 6-digit level, specific to a country and period, we define a "similar group" consisting of all products with the same 4-digit code. Within this group, we calculate the minimum tariff exposure. These similar products are subject to lower taxes than the correct classification, incentivizing importers to misclassify products to evade customs duties. In essence, maintaining the tariff rate for a particular product while taking advantage of lower tariffs on similar products can lead to increased levels of evasion. To address our conceptual framework we consider the following econometric specifications:

$$y_{cpt} = \beta_1 T_{cpt} + \beta_2 T_{cpt}^{sim} + \lambda' X_{cpt} + \delta_{cp} + \delta_{ct} + \delta_{pt} + \varepsilon_{cpt}, \tag{6}$$

Where y refers to the trade gap (our evasion measures for values and quantities), $t \in \{2003, 2004\}$ reflects the PIS and COFINS shock when only 2004 has been affected.[35] The total tariff exposure to which goods are subject is denoted as T_{cpt}, and it is contingent upon the originating country due to Brazil's significant trade agreement, MERCOSUR. We quantify tariff exposure as the average applied tariff rates for each transaction within the Administrative sample. T_{cpt}^{sim} represents the minimum tariff exposure for similar products. Additionally, we introduce fixed effects for pairs of country-product δ_{cp}, country-year δ_{ct}, and product-year δ_{pt} to account for unobservable factors associated with these pairings.[36]

In our main specification, we also incorporate control variables and administrative treatments

[34]The machine learning analysis, as detailed in Appendix 4, concentrates on elucidating the variations in evasion by examining observable import characteristics.

[35]Another interpretation of our empirical strategy is similar to a shift-share approach with only one industry. In that case, we must assume that the units that received stronger shocks would have the same evolution of potential outcomes as those that received weaker tax shocks.

[36]The product-time fixed effects capture changes in the productivity of a specific product shared among countries and shifts in trade policies targeting specific products, whereas the country-product fixed effects control for a country's comparative advantage in exporting a particular product p. The country-year pairings may address changes in the productivity of an exporting country over time.

that could confound the relationship between tariff exposure and evasion, denoted as X_{cpt}. These administrative treatments include exemption status from PIS and COFINS, tariff agreements (i.e., MERCOSUR), anti-dumping duties, and drawbacks. Additionally, our control variables encompass the average exchange rate, the proportion of trading companies, the proportion of audits, the Brazilian states where imports were cleared, average clearance time, insurance and freight costs, the number of imported products, the number of unique importers, and the years of activity for the importer.[37]

The variation in tariffs induced by the shock is due to the incidence nature of PIS and COFINS taxes.[38] Therefore, the variation in customs duties burden faced by importers due to the introduction of those two taxes on imports of goods is proportional to the pre-shock total tariff rate exposure, $(1 + II_{cp}) \times (1 + IPI_{cp})$. Since this exposure is set on a product basis, the shock created variation across products even though PIS and COFINS rates were generally the same for all products.[39]

3.3 NTBs and their influence on evasion responses

Non-tariff barriers have gained significant prominence as a trade policy tool in recent decades, as highlighted by the World Trade Organization (WTO 2012). Grübler and Reiter (2021) has documented the evolution of NTM (Non-Tariff Measure) notifications to the WTO, revealing a substantial increase from approximately one thousand notifications in 1996 to approximately four thousand notifications in 2019 (Grübler and Reiter 2021, Figure 8, p.p. 149).

We incorporate NTB into our analysis and its interaction with our tariff exposure variable in Equation (6). This results in the following econometric specification:

$$y_{cpt} = \beta_1 T_{cpt} + \beta_2 T_{cpt} \cdot NTB_{cpt} + \beta_3 T^{sim}_{cpt} + \beta_4 NTB_{cpt} + \lambda' X_{cpt} + \delta_c + \delta_p + \delta_t + \varepsilon_{cpt} \quad (7)$$

Where y_{cpt} assumes one of the four evasion measures as in the equation (6) and, NTB_{cpt} is the proportion of product p imports, coming from partner country c at year t subject to import licensing, our representative NTB. In addition, we include fixed effects without interaction. Since the NTB designation is associated with a *per* product basis, the product-year fixed effect would absorb most of the effect we are interested in. We are mostly interested in estimating the parameter β_2, i.e., the modulating effect of NTB on misreporting responses.

4 Results

We begin by discussing the results of the elasticity of customs duties evasion (ECDE) analysis. Next, we examine the influence of Non-Tariff Barriers as an ECDE modulator. Finally, we present the heterogeneous results across firms, products, and Brazil's trade bloc partners.

[37] Appendix 3 provides additional robustness checks for this linear specification. Table 3.5 begins with the main specification and incrementally includes the average exchange rate and the proportion of import value by state, re-estimating the model. The results do not qualitatively change.

[38] $\Delta T_{cp} = T_{cp,2004} - T_{cp,2003} = (Pis + COFINS) \times (1 + II_{cp}) \times (1 + IPI_{cp})$.

[39] Considering the wide PIS and COFINS tax base across products and origin country, endogeneity issues will not likely arise. Even though, for each set of results, we provide several robustness assessments and falsification tests in the Appendix 3.

4.1 ECDE

Table 3 summarizes our first set of estimation results for the ECDE. We estimate Equation (6) while considering two categories of tariffs and four different evasion measures, resulting in eight models estimated.

In the first set of models, columns 1-4, the tariff is calculated using a straightforward average of the effectively applied tariff rates, a metric available exclusively in the administrative dataset. [40] The subsequent set of models, columns 5-8, employs the Most Favored Nation (MFN) tariff, a metric utilized in studies that rely on data from WITS, coming from our *trade sample*.

The estimation results exhibit notable disparities between the two data sources. We observe higher and statistically significant elasticities when the administrative data is employed (first four columns). For instance, the estimated value of ECDE using the administrative data is 1.98, indicating that for a one percentage point increase in total tariff exposure, evasion rises by 1.98%. Conversely, when the regressions are conducted using the trade data (MFN tariffs), the resulting elasticities tend to be lower, accompanied by larger standard errors. Notably, while the results for evasion involving values are estimated with a much smaller magnitude (1.12), the associated estimates for evasion quantities are not statistically significant at the 10% level. Moreover, the coefficient for similar product tariffs is no longer statistically significant for MFN tariff models.

[40] We also conduct regressions using FOB (Free on Board) value-weighted tariffs. Both methods of aggregation have their advantages and drawbacks. While the weighted average tariff may raise concerns about endogeneity since our dependent variable relies on FOB values, the simple average gives equal weight to products with different import value shares but the same number of transactions. The results for the weighted average tariffs exhibit similar qualitative findings and are available upon request.

Table 3: Regression results comparing effectively applied against MFN tariff rates with the Placebo Regression

Panel A: Main Estimates

	Administrative Sample - Effectively Applied Tariff				Trade Sample - MFN Tariff			
Dep. Var.:	Value	Ext. Value	Quantity	Ext. Quantity	Value	Ext. Value	Quantity	Ext. Quantity
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Tariff	1.9840***	1.5444***	0.9657**	0.8679**	1.1282**	1.0570***	-0.1171	-0.1856
	(0.3082)	(0.2382)	(0.4401)	(0.4059)	(0.5614)	(0.3687)	(0.6897)	(0.5927)
Similar	-0.4960**	-0.3894**	-0.2878	-0.2546	-0.0314	-0.0440	-0.0067	0.0075
	(0.2131)	(0.1494)	(0.3208)	(0.2875)	(0.2264)	(0.1431)	(0.3168)	(0.2803)
Num.Obs.	63226	64481	57267	59023	63226	64481	57267	59023
R2 Adj.	0.538	0.570	0.546	0.619	0.537	0.569	0.545	0.619

Panel B: Validity Check

	Matched Sample: 2003-2004				Placebo:2001-2002			
Dep. Var.:	Value	Ext. Value	Quantity	Ext. Quantity	Value	Ext. Value	Quantity	Ext. Quantity
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Tariff	1.7152***	1.4299***	0.7151	0.6640	0.4499	0.2317	-0.2021	0.0550
	(0.4371)	(0.3273)	(0.5087)	(0.4728)	(0.4268)	(0.3554)	(0.4814)	(0.4816)
Similar	-0.6331***	-0.4813***	-0.3154	-0.3293	-0.1074	-0.0568	-0.0500	-0.0687
	(0.2207)	(0.1525)	(0.3166)	(0.3048)	(0.2493)	(0.2053)	(0.2982)	(0.2961)
Num.Obs.	46379	47351	42308	43366	45405	46334	41555	42460
R2 Adj.	0.515	0.593	0.523	0.626	0.472	0.540	0.454	0.567
FE: cp	✓	✓	✓	✓	✓	✓	✓	✓
FE: ct	✓	✓	✓	✓	✓	✓	✓	✓
FE: pt	✓	✓	✓	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Panel A: Tariff is calculated as a simple average from transaction-level data by origin country, year and 6-digit product or the most favored nation accordingly. All standard errors are two-way clustered at country and product. All regressions include additional control variables. Tariffs on similar products and their own are calculated as simple averages from transaction-level data by origin country, year, and 6-digit product. All standard errors are two-way clustered at country and product. All regressions include additional control variables. Panel B: We include only observations where product-country is present in all four years, 2001 through 2004. Matched Sample results are those for the years 2003 and 2004, while Placebo results use trade gaps taken from the years 2001 and 2002.

This discrepancy in results with the MFN tariff is not unexpected, as importers typically concern themselves with the *effectively* applied tariff rates on their imported goods, which only the administrative data captures appropriately. Therefore, the findings presented in Table 3 reinforce our assertion that administrative data should be the preferred choice for assessing evasion whenever available, particularly in the context of developing economies where customs duties extend beyond the scope of the MFN tariff itself.[41]

Our findings of statistically significant quantity evasion responses to tariffs stand in contrast to previous literature (e.g., Javorcik and Narciso (2008) and Fisman and Wei (2004)), particularly when not controlling for similar products. These results in columns 7-8 suggest that the MFN-tariff (trade) data may not comprehensively capture the complexities of tariff evasion behaviors within the specific context of a developing nation that employs many simultaneous trade policies. In Brazil's case, several factors can contribute to the observed quantity evasion responses. The tight regulation of prices through the Customs Valuation Agreement (CVA), combined with high effective tariff rates (e.g., Import Duty, Industrialized Products Tax, PIS, and COFINS), may incentivize importers to engage in quantity evasion (i.e., smuggling). This aligns with the findings of Javorcik and Narciso (2017), who suggest that this combination of factors can trigger such behavior. While customs officials may detect smuggling by comparing reported and actual shipment container weights, the interaction of strict price controls and high tariff rates creates an economic incentive for importers to evade taxes by underreporting quantities. This nuanced context in Brazil may explain the observed quantity evasion responses, which differ from previous findings in other settings.

Table 3 also provides evidence that the tariff rate on similar products substantially affects evasion in value terms.[42] Notably, when considering the minimum tariff rate among products sharing the same 4-digit classification, the coefficients for the $Tariff$ variable are slightly larger, yet they remain statistically significant at the 1% level. Concurrently, the coefficients associated with similar products maintain the expected negative sign. These findings suggest that importers tend to misreport their high-taxed goods as low-taxed varieties, opting for the classification that incurs the lowest tariff within a group of similar HS codes.

Validity check. We conducted a placebo-like regression exercise using trade gap data from 2001 and 2002 to assess the validity of the parallel trends assumption in our differences-in-differences identification strategy in Table 3. We restricted our dataset to observations where the product-partner was present in all four years from 2001 to 2004. Therefore, we rerun the regression in Panel A but using our restricted sample with the results shown in Panel B, columns 1-4. Our placebo model, shown in Panel B, columns 5-8, replaced the 2003/04 trade gap for a given product originating from a specific country with the corresponding 2001/02 trade gap, and two key observations stand out.

First, in columns 1-4, we observe that the estimated coefficients are largely consistent with our findings from the full sample as presented in Panel A Table 3. Second, the previously statistically

[41]For illustration purposes, we provide in Table 2.7 (Appendix 2) the difference of averages in the effectively applied tariff rate and the most favored nation tariff rate for the top five most imported products (HS 6-digit) in 2003 and 2004.

[42]We have also conducted several robustness checks, such as considering the average tariff rate of similar products (excluding the product at hand) rather than the minimum rate, as well as assessing the impact of both simple and weighted averages of similar products on evasion (detailed in Table 3.1, Appendix 3). The results from these robustness checks are available in Appendix 3.

significant coefficients associated with the *Tariff* and *Similar* explanatory variables lose significance in the placebo analysis. This is due to a reduction in the estimated effect size across all models, while the standard errors do not show substantial changes. These findings reinforce the robustness of our empirical strategy by highlighting that the results are not driven by spurious trends or anomalies in the data, thus providing further support for the validity of our differences-in-differences approach.

Robustness Checks. We run a set of models starting from a baseline specification and incrementally including covariates to obtain our full specification shown in Table 4 (Administrative Sample). The baseline includes only tariff exposure and the minimum tariff of similar products. We add the tariff agreement, anti-dumping, drawback, PIS/COFINS exemption, audits, average exchange rate, proportion of trading companies, average insurance and freight costs, number of imports and importers, average years of activity for importers, and average days until clearance to arrive at the administrative specification. Lastly, the State specification includes the proportion of imports made through each Brazilian State.[43] Additionally, we explore the possibility of a nonlinear effect of tax rates on customs duty evasion (Fisman and Wei 2004; Dowd, Landefeld, and Moore 2017; Fuest, Hugger, and Neumeier 2022) by introducing a squared tariff exposure term into our regression specified in equation (6), at Table 2.2 (Appendix 2).

[43]The state of Tocantins did not record any imports in both periods. São Paulo is chosen as the reference State.

Table 4: Robustness assessment to covariates inclusion

	Baseline				Administrative				States			
Dep. Var.:	Value	Ext. Value	Quantity	Ext. Quantity	Value	Ext. Value	Quantity	Ext. Quantity	Value	Ext. Value	Quantity	Ext. Quantity
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Tariff	2.43***	1.92***	1.45***	1.31***	2.04***	1.54***	0.92**	0.79*	1.98***	1.54***	0.97**	0.87**
	(0.32)	(0.23)	(0.42)	(0.38)	(0.33)	(0.24)	(0.44)	(0.40)	(0.31)	(0.24)	(0.44)	(0.41)
Similar	-0.54**	-0.42***	-0.36	-0.35	-0.51**	-0.40***	-0.31	-0.29	-0.50**	-0.39**	-0.29	-0.25
	(0.22)	(0.16)	(0.33)	(0.29)	(0.21)	(0.15)	(0.32)	(0.29)	(0.21)	(0.15)	(0.32)	(0.29)
Num.Obs.	63458	64715	57457	59227	63226	64481	57267	59023	63226	64481	57267	59023
R2 Adj.	0.532	0.565	0.543	0.616	0.536	0.568	0.545	0.618	0.538	0.570	0.546	0.619
FE: cp	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
FE: ct	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
FE: pt	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The baseline specification include as regressors only the tariff exposure and the minimum tariff of similar products. The administrative specification is incremental, adding tariff agreement, anti-dumping, drawback, PIS/COFINS exemption, audits, exchange rate, tradings, insurance and freight cost, number of imports and importers and the average company's years in activity and days until the import is cleared. States specification additionally include the proportion of imports made through each Brazilian State. Tariffs on similar products and its own are calculated as a simple average from transaction-level data by origin country, year and 6-digit product. Similar products have the same first 4-digits. All standard errors are two-way clustered at country and product.

4.2 The NTB effect

This section investigates how the imposition of our NTB restriction influences the elasticity of customs duty evasion.[44] We estimate equation (7) and present the main regression coefficients for the effects of NTBs in Table 5 below:

Table 5: Effect of NTB on unreported trade

Dep. Var.:	Value	Ext. Value	Quantity	Ext. Quantity
	(1)	(2)	(3)	(4)
Tariff	2.6194***	2.1622***	1.8979***	1.8751***
	(0.2849)	(0.1995)	(0.3824)	(0.3573)
Tariff X NTB	-1.8592***	-1.4200***	-1.9916***	-1.8288***
	(0.3943)	(0.2722)	(0.4874)	(0.4607)
Similar	-0.6280***	-0.6048***	-0.7158**	-0.6927**
	(0.2281)	(0.1577)	(0.2837)	(0.2723)
NTB	0.0250	-0.0180	0.2338*	0.1697
	(0.0933)	(0.0652)	(0.1382)	(0.1261)
Num.Obs.	63226	64481	57267	59023
Avg. Tariff	0.1908	0.1921	0.1894	0.1909
Avg. NTB	0.2647	0.2646	0.2696	0.2684
R2 Adj.	0.121	0.187	0.185	0.311
FE: c	✓	✓	✓	✓
FE: p	✓	✓	✓	✓
FE: t	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Tariff is calculated as a simple average from transaction-level data by origin country, year and 6-digit product from the administrative dataset. NTB is the proportion of imports subject to non-automatic licensing for a given triplet, country, product, and year. All standard errors are two-way clustered at country and product. All regressions include additional control variables.

The imposition of an import license modulates the elasticity of evasion, even after controlling for tariff rates and misclassification on similar products, consonant with our machine learning model's suggestion in Appendix 4 and conceptual framework.[45] For every evasion measure in Table 5, we find that the higher the proportion of imports subject to the NTB, the lower the elasticity of tariff evasion, i.e., although the imposition of an import license does not affect the changes in the unreported trade directly, it modulates the elasticity of tariff evasion (Tariff X NTB coefficient). Our results indicate that a higher proportion of imports subject to the NTB corresponds to a

[44] We check whether the proportion of transactions involving non-tariff barriers has significantly changed from the pre to post-shock period. Figure 3.5 (Appendix 3) shows the distribution of NTBs by tariff quintile for 2003 and 2004. The chart shows that NTB-related imports did not change their distribution across tariff quintiles.

[45] Our machine learning model (Appendix 4) shows that NTBs consistently ranks among the top three most significant features for this elasticity heterogeneity.

lower elasticity of tariff evasion, which suggests that import licensing restrictions constrain the responsiveness of evasion behavior to changes in customs duties.[46]

Our Table 5 presents the estimated coefficients with the marginal effect of tariffs on evasion at the mean point of NTB, which is calculated as 2.13 (2.62 - 1.86 * 0.265), larger than 1.98 (Table 3, Panel A, model 1). However, the estimated elasticity response decreases to 0.76 (2.62 - 1.86 * 1) with all imported goods subject to the NTB. For quantity evasion, we find similar qualitative results. Still, the corresponding estimated elasticity would be negative (-0.09 = 1.9-1.99) if all imported goods are subject to the NTB (column (3) of Table 5). These highlight the importance of NTBs in shaping evasion dynamics.

Non-tariff barriers serve as a substitute for tariff reduction in mitigating customs duties evasion (Kee and Nicita (2016)). Import licensing restrictions can effectively deter evasion, potentially reducing the need for lower tariff rates to achieve the same outcome. This effect appears to be associated with improved enforcement quality facilitated by access to detailed information where filing for an import license creates a verifiable paper trail for importers, which can be later utilized by auditors (Pomeranz 2015; Kopczuk and Slemrod 2006; Gordon and Li 2009; Kleven et al. 2011; Arbex and Mattos 2015; Slemrod et al. 2017; Naritomi 2019).[47] The mitigation of unreported trade resulting from non-tariff barriers appears to be an unintended consequence, as NTBs are not designed specifically to curb evasive behavior. Instead, the effectiveness of NTBs in reducing unreported trade is likely due to their role in enhancing enforcement quality through collecting and verifying specific import information (importer identification, country of origin, clearance port, exporter details, and specific information about the imported goods, such as the NCM classification, administrative treatments, taxation regime, payment method, and unit value of the imported product).[48]

Robustness Checks. Our Figure 3 shows our baseline specification and models that incrementally include covariates to obtain our full specification model. The covariates included in the baseline, administrative, and state specifications are the same as described in Table 4 footnote. We observe that adding controls does not change our results.[49]

[46]We computed a permutation test on the interaction term $Tariff \times NTB$. The results are available upon request.

[47]The observed lower elasticity of unreported trade for products subject to non-tariff barriers (NTBs), even at similar tariff levels, seems not primarily due to increased audits on NTB-subject imports. We can see this in Table 3.4 (Appendix 3), which provides a balance table on imports subject to NTBs.

[48]The negative sign associated with the interaction of tariff rate and the proportion of NTBs is consistent with our theoretical framework, which posits that the quality of enforcement reduces the elasticity of evasion. It is essential to note that the tariff equivalent of NTBs on imports differs from their tariff rate. The imposition of an NTB would effectively raise the tariff equivalent. We simulated the result of a 20% increase in tariffs for NTB-subject imports, and these results are presented in Table 3.9 in Appendix 3.

[49]See also the Table 3.5 (Appendix 3).

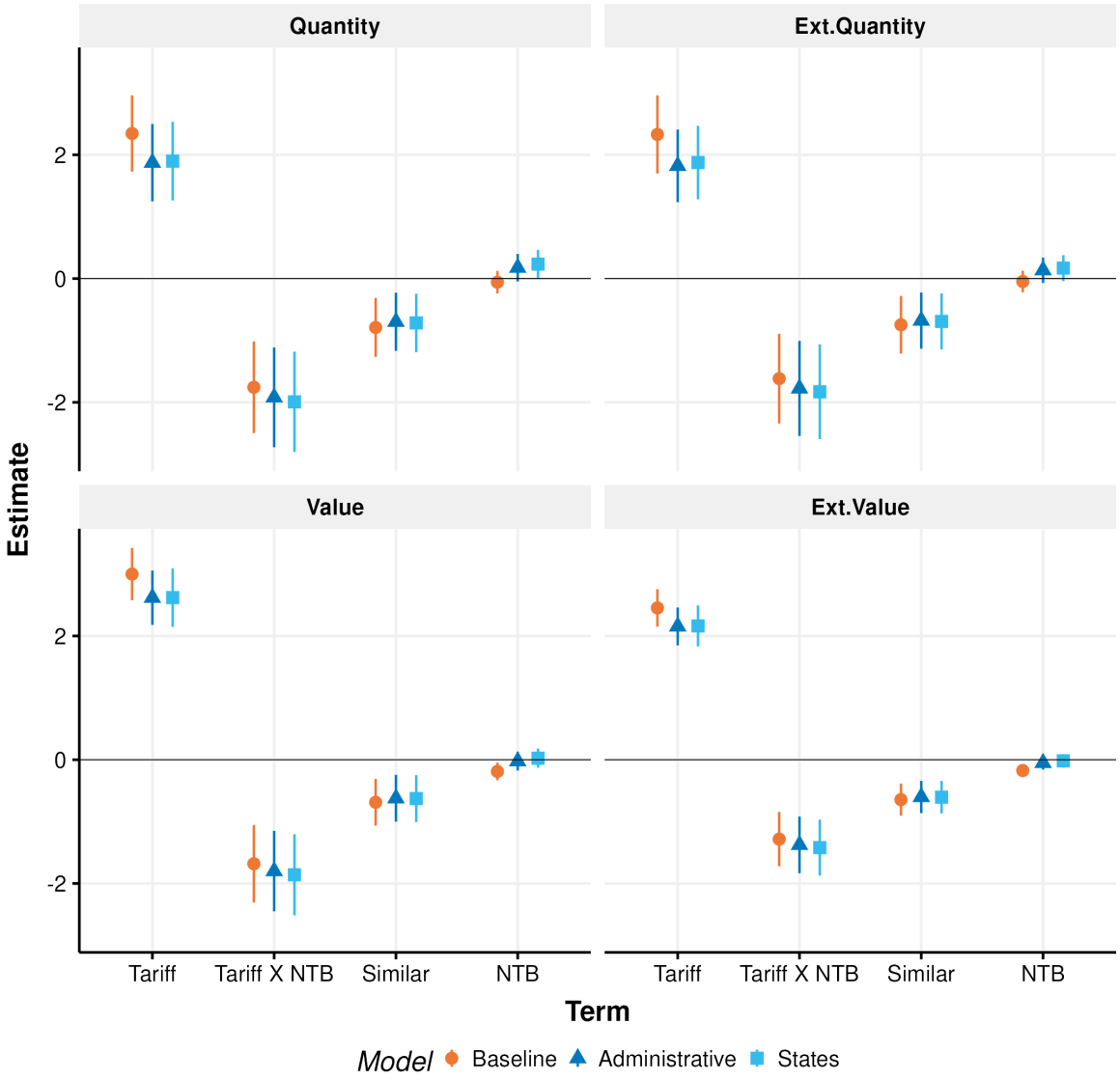


Figure 3: Robustness assessment for different specifications of NTB effect on evasion elasticity. The figure displays the estimated coefficients based on equation 7 and their corresponding 95% confidence intervals. The baseline model includes only the minimum tariff exposure and NTBs (level) of similar products, other than the fixed effects described in Equation 7. The administrative model adds tariff agreement, anti-dumping, drawback, PIS/COFINS exemption, audits, average exchange rate, the proportion of trading companies, average insurance and freight costs, number of imports and importers, average importer's years in activity, and average days until clearance. The State's model includes the proportion of imports made through each Brazilian State. See these results in Table 3.3 (Appendix 3).

Last, to address the concern that the NTB interaction specification might capture a dosage effect of tariffs on evasion, we examine the possibility that NTBs are concentrated mainly on low-tariff products. We replace the proportion of NTBs with a low-tariff indicator (Low Tariff), a dummy variable that identifies products with tariffs below the median. The results (Table 3.7 in Appendix 3) offer reassurance that our NTB modulator effect on evasion is not merely a consequence of its correlation with low tariffs. Next, we investigate the heterogeneous effect of NTBs on unreported

trade.

4.3 Heterogeneity Results

Firms Heterogeneity. We categorize firms based on their activity classification code (CNAE) into two groups: industry/manufacturing or non-manufacturing. This allows us to measure the proportion of manufacturing firms for each product-country-year unit.

First, we observe that the share of manufacturing firms is around 50% in both years of our sample. This share does not vary significantly across the four measures of unreported trade. Second, the regression results presented in Table 6 highlight the significant interaction effects between tariffs and non-tariff barriers (NTBs) on unreported trade values and quantities. Specifically, the coefficient of the interaction term $Tariff \times NTB$ is consistently negative and statistically significant across all specifications, indicating that the presence of NTBs weakens the positive impact of tariffs on trade outcomes. For instance, in column (1), the coefficient is -1.55, significant at the 1% level, suggesting that NTBs mitigate the tariff-induced increase in unreported trade to 1.1 (2.65-1.55). This pattern persists across the various specifications, reinforcing the idea that NTBs introduce additional frictions that counteract the trade-enhancing effects of tariff measures. Moreover, the magnitude of this interaction effect varies across different dependent variables, ranging close to -1 for quantity-based measures, implying that NTBs directly reduce unreported trade value responses to tariffs but eliminate unreported trade volumes (for instance, column 12 shows 0.98-0.93 = 0.05).

Table 6: Regression results of interacting NTB and proportion of manufacturers.

	Baseline				Administrative				States			
Dep. Var.:	Value	Ext. Value	Quantity	Ext. Quantity	Value	Ext. Value	Quantity	Ext. Quantity	Value	Ext. Value	Quantity	Ext. Quantity
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Tariff	2.65***	2.21***	1.25**	1.27***	2.30***	1.93***	0.91*	0.89*	2.34***	1.96***	0.98*	0.98**
	(0.33)	(0.23)	(0.50)	(0.48)	(0.35)	(0.23)	(0.50)	(0.48)	(0.36)	(0.24)	(0.50)	(0.48)
Tariff X NTB	-1.55***	-1.01***	-1.14**	-1.01**	-1.41***	-0.93***	-0.98*	-0.85*	-1.46***	-0.97***	-1.07**	-0.93*
	(0.42)	(0.30)	(0.48)	(0.47)	(0.42)	(0.30)	(0.50)	(0.49)	(0.43)	(0.30)	(0.51)	(0.50)
Similar	-0.71***	-0.66***	-0.72**	-0.69**	-0.64***	-0.62***	-0.64**	-0.63**	-0.65***	-0.62***	-0.66**	-0.65**
	(0.23)	(0.16)	(0.31)	(0.30)	(0.23)	(0.16)	(0.30)	(0.29)	(0.24)	(0.16)	(0.30)	(0.29)
NTB	-0.17*	-0.19***	-0.14	-0.11	-0.14	-0.16***	-0.08	-0.10	-0.09	-0.13**	0.00	-0.04
	(0.09)	(0.06)	(0.13)	(0.12)	(0.09)	(0.06)	(0.13)	(0.12)	(0.09)	(0.06)	(0.14)	(0.13)
Tariff X NTB X Manufacture	-0.09	-0.53	-0.98	-0.94	-0.57	-0.81	-1.57*	-1.54*	-0.59	-0.82*	-1.55*	-1.52*
	(0.61)	(0.47)	(0.85)	(0.81)	(0.62)	(0.49)	(0.92)	(0.83)	(0.62)	(0.49)	(0.92)	(0.84)
Avg. Tariff	0.19	0.19	0.19	0.19	0.19	0.19	0.19	0.19	0.19	0.19	0.19	0.19
Avg. NTB	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27
Num.Obs.	59732	60908	54125	55784	59513	60687	53944	55590	59513	60687	53944	55590
R2 Adj.	0.115	0.181	0.181	0.307	0.119	0.186	0.186	0.312	0.120	0.187	0.188	0.314
FE: c	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
FE: p	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
FE: t	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Results of linear regressions interacting tariffs, NTB and proportion of imports from manufacturing companies. The administrative dataset is used. Two-way clustered (product and country) standard erros in parentheses. All pairwise interactions of Tariff, NTB and Manufacturer are included in the model, although not shown due to space considerations.

Additionally, the three-way interaction term *Tariff* × *NTB* × *Manufacture* presents a more nuanced dynamic. Although consistently negative across all specifications, it exhibits a statistically significant coefficient, particularly in columns (9) to (12) when all control variables are included. For instance, in column (11), the coefficient is -1.55, significant at the 10% level, suggesting that the adverse effects of NTBs on tariff-induced unreported trade responses are exacerbated in the context of manufacturing-intensive sectors. This result implies that manufacturing firms, typically more sensitive to trade restrictions, face heightened barriers to misreport trade when tariffs and NTBs are in place. In this case, we find an elasticity of unreported trade to tariff close to -1.64 (0.98-1.07-1.55)[50], i.e, after a 1% increase in tariff, a manufacturing firm with all the transactions under NTBs would reduce unreported trade by 1.64%.[51]

Discussion. Manufacturing firms tend to import a broader range of intermediate and capital goods, often subject to detailed regulatory requirements, such as technical standards and product certifications. This makes them more exposed to regulation, not just in terms of trade value, but also in terms of the quantity, quality, and safety of imported inputs. As a result, manufacturers are more likely to respond to NTBs by adjusting their documentation and compliance practices to avoid disruptions. Empirical studies have shown that NTBs disproportionately affect sectors characterized by high input intensity and production complexity, both of which are typical of manufacturing activities (Gasiorek, Holmes, and Rollo 2012). In this context, NTBs act not just as trade frictions but also as implicit enforcement mechanisms, especially for firms that operate across multiple borders and are more sensitive to compliance costs and shipment delays (Cadot, Gourdon, and Tongeren 2015).

Product Heterogeneity. Next, we also explore the heterogeneity of a product based on the Broad Economic Categories (BEC) classification: Consumption-input (C-I), consumption (C), capital (K), and input (I) goods. Table 7 presents descriptive statistics for their categorized products with a larger share of inputs (64,142) compared to other products, showing the proportion of manufacturing importers and the average Non-Tariff Barriers (NTBs) across different product categories. Capital goods and inputs exhibit the highest share of manufacturing importers, with averages of 0.4955 and 0.5589, respectively, suggesting that these product categories are more relevant for industrial use. On the other hand, consumption and consumption-input goods show lower averages of 0.2474 and 0.1905, respectively, indicating a smaller role for manufacturing firms in these product categories.

Regarding NTBs, consumption goods (C) display the highest average NTB (0.3443), followed by inputs (I) at 0.2478. In contrast, consumption-input goods (C-I) show a lower average of 0.2329 and capital goods (K) have the lowest NTB average of 0.2203. The standard deviations suggest considerable heterogeneity in both manufacturing importer shares and NTB levels within each category. Furthermore, the correlation between the average share of manufacturing importers and NTB levels is -0.42, indicating a negative relationship—categories with a higher presence of manufacturing firms tend to face lower NTBs on average. These findings highlight the distinct trade patterns and regulatory environments faced by different types of goods under the BEC classification.

[50]This is a simplification since we are assuming other terms involving the Tariff are zero.

[51]We have very detailed information about importers, and we can identify trading companies operating in the country. Trading companies may serve as intermediaries with specialization in international trade and presumably have a body of experts on their staff. We show in Appendix 5 that even though trading companies are supposed to better understand customs duties legislation, they seem to have a similar unreported trade response to tariff changes than the ordinary importer.

Table 7: BEC products descriptive statistics

BEC Product	Share Manufacture	Std. Dev.	Share NTB	Std. Dev.	N
C-I	0.1905	0.3356	0.2329	0.3961	11297
C	0.2474	0.3597	0.3443	0.4325	13789
K	0.4955	0.3993	0.2203	0.3352	17646
I	0.5589	0.4052	0.2478	0.3756	64142

Notes: This table shows the proportion of manufacturing importers and NTB's by BEC Product (with standard deviation) and number of observations. The correlation of average manufacturing and average NTB is -0.42. BEC codes are C: consumption, C-I: consumption-input, I: input, K: capital.

Our regression in Table 8 examines the interaction between tariffs, non-tariff barriers (NTBs), and Broad Economic Categories (BEC). Unlike our previous section, the interaction term between tariffs and NTBs is not statistically significant across specifications, suggesting that the combined effect of these trade barriers does not systematically influence the unreported trade of BEC-categorized products. This is the case because the coefficient of the interaction between Tariff and NTB now captures the impact on the omitted category product (consumption good). Substantial differences emerge in the interaction of other BEC categories. Columns (7-8) and (11-12) show the lack of unreported quantity responses to tariffs. This relationship with the tariff is only observed with the triple Tariff $\times$ NTB $\times$ Input (I) and Capital (K) coefficients, which are negative and statistically significant. This suggests that higher tariffs combined with NTBs lead to a statistically significant reduction in unreported trade values and quantities for input goods (-2.25; -3.35. columns 9 and 11, respectively) and capital goods compared to the baseline product (consumption). The negative effect is even more substantial for capital goods, where the coefficients are consistently larger in magnitude and highly significant across all specifications (-3.43; -3.51. columns 9 and 11, respectively).

Table 8: Regression results of interacting NTB and BEC products.

	Baseline				Administrative				States			
Dep. Var.:	Value	Ext. Value	Quantity	Ext. Quantity	Value	Ext. Value	Quantity	Ext. Quantity	Value	Ext. Value	Quantity	Ext. Quantity
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Tariff	1.95***	1.46***	1.26*	1.04	1.75***	1.35***	1.12	0.93	1.81***	1.40***	1.22	1.05
	(0.54)	(0.43)	(0.71)	(0.68)	(0.55)	(0.42)	(0.73)	(0.69)	(0.55)	(0.41)	(0.75)	(0.70)
Tariff X NTB	-0.15	-0.27	0.20	0.21	-0.19	-0.40	0.11	0.05	-0.31	-0.46	-0.05	-0.07
	(0.61)	(0.35)	(0.80)	(0.73)	(0.63)	(0.36)	(0.82)	(0.74)	(0.62)	(0.35)	(0.82)	(0.74)
Similar	-0.71***	-0.67***	-0.74**	-0.70**	-0.65***	-0.62***	-0.65**	-0.64**	-0.65***	-0.63***	-0.67**	-0.66**
	(0.23)	(0.16)	(0.31)	(0.30)	(0.24)	(0.16)	(0.30)	(0.29)	(0.24)	(0.16)	(0.30)	(0.29)
NTB	-0.29	-0.20	-0.45	-0.38	-0.23	-0.13	-0.29	-0.22	-0.19	-0.11	-0.25	-0.21
	(0.25)	(0.15)	(0.30)	(0.26)	(0.25)	(0.15)	(0.31)	(0.27)	(0.25)	(0.15)	(0.30)	(0.26)
Tariff X NTB X C-I	-0.51	0.06	0.41	0.79	-0.54	0.16	0.23	0.65	-0.45	0.21	0.27	0.66
	(1.01)	(0.65)	(1.33)	(1.26)	(0.99)	(0.62)	(1.32)	(1.25)	(1.00)	(0.61)	(1.35)	(1.26)
Tariff X NTB X I	-2.25**	-1.31**	-3.40***	-3.02***	-2.37***	-1.26**	-3.52***	-2.99***	-2.25**	-1.20**	-3.35***	-2.84***
	(0.86)	(0.54)	(1.14)	(1.05)	(0.87)	(0.53)	(1.15)	(1.05)	(0.86)	(0.52)	(1.16)	(1.06)
Tariff X NTB X K	-3.63***	-2.94***	-3.83***	-3.62***	-3.46***	-2.69***	-3.66***	-3.67***	-3.43***	-2.71***	-3.51***	-3.56***
	(1.14)	(0.82)	(1.13)	(1.04)	(1.24)	(0.85)	(1.23)	(1.17)	(1.25)	(0.86)	(1.25)	(1.18)
Avg. Tariff	0.19	0.19	0.19	0.19	0.19	0.19	0.19	0.19	0.19	0.19	0.19	0.19
Avg. NTB	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27
Num.Obs.	59732	60908	54125	55784	59513	60687	53944	55590	59513	60687	53944	55590
R2 Adj.	0.114	0.180	0.181	0.307	0.119	0.186	0.186	0.312	0.120	0.187	0.188	0.314
FE: c	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
FE: p	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
FE: t	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Results of linear regressions interacting tariffs, NTB, and BEC products. The reference product is 'Consumption'. Code 'C-I' refers to consumption-input products, and 'K' refers to capital goods. The administrative dataset is used. Two-way clustered (product and country) standard errors are in parentheses. All pairwise interactions of Tariff, NTB, and BEC codes are included in the model, although not shown due to space considerations.

The results highlight heterogeneity in the response of different product categories to trade barriers. While the interaction term for Tariff × NTB × C-I is not statistically significant, those for inputs (I) and capital (K) goods show strong adverse effects compared to the consumption goods (baseline). This implies that consumption-intermediary goods do not face the same degree of unreported trade suppression from combined tariff and NTB measures, likely because these goods react similarly to the baseline products. In contrast, inputs and capital goods are affected, as they often require specific inputs, regulatory approvals, or long-term contracts that make them more traceable. The magnitude of the effects is particularly large for capital goods, where the interaction term remains significant at the 1% level across all regression specifications. This suggests that policy measures affecting trade barriers in capital goods markets can disproportionately affect unreported trade flows.[52]

Discussion. A plausible explanation for the differential response to NTBs lies in the characteristics of the transactions involved in importing inputs and capital goods, which are typically operated within structured supply chains and must comply with stricter regulatory and reporting standards. Accurate documentation is often required to access input tax credits, duty drawback schemes, or to qualify for special customs regimes, reducing the incentive and opportunity to underreport (Yang 2008; Javorcik and Narciso 2008). In contrast, consumption goods are more commonly imported by smaller or less formal firms in the retail and distribution sectors. These importers are subject to less oversight and regulatory pressure, giving them greater flexibility to engage in evasion or misreporting practices (Fisman and Wei 2004).

Trade-block origin of the product. Last, we move to understand the heterogeneous impact of unreported trade responses based on the origin of the imports. Our Table 9 presents the statistics for the following trade blocs: Mercosur (Brazil, Argentina, Paraguay, Uruguay), USMCA (U.S., Canada, Mexico), the EU (European Union member states), and China, compared to all other trading partners. We run a simple linear regression of the proportion of NTBs and tariff exposure on trade-blocks, where "Other" is the reference. The estimates indicate that imports from "Other" countries (outside the specified blocs) exhibit about 27% of their imports associated with NTBs and 20% on average for tariff exposure. Those are the baseline results, and all other block coefficients are measured in addition to the baseline. MERCOSUR countries, on the other hand, display the highest NTB association (0.2947=0.2710+0.0237) and the lowest tariff exposure (0.1089=0.2034-0.0945), reinforcing the reduced tariff applied for imports within this bloc. The USMCA bloc shows no significant differential relationship with NTBs and a small but significant negative coefficient for tariff exposure (-0.0165), implying that tariff rates for these imports tend to be lower. For EU imports, the results reveal a negative and significant relationship with NTBs (-0.0454), indicating that European imports are less likely to be exposed to non-tariff barriers. However, tariff exposure for EU imports is slightly higher than the baseline, and this difference is statistically significant ($p = 0.0017$). Chinese imports exhibit an insignificant differential relationship with NTBs, with a small, positive, and significant coefficient for tariff exposure (0.0053), indicating a slightly higher association with tariffs. Overall, the regression results highlight the heterogeneous association of trade policies across different trade partners.

[52]We also explore enforcement quality heterogeneity based on specific attributes of the imported product. We follow the classification system of Rauch (1999) for differentiated goods, based on their trade patterns and market structure. We show the results in Appendix 5, where the three-way interaction term (Tariff × NTB × Differentiated Products) is not statistically significant across all specifications, indicating that the impact of tariffs and NTBs does not significantly differ between differentiated and non-differentiated products under the Rauch (1999) classification.

Table 9: Descriptive regression results of NTBs and average tariff exposure on trade blocs.

	NTB (1)	Tariff (2)
Other	0.2710***	0.2034***
	(0.0019)	(0.0006)
MERCOSUR	0.0237***	-0.0945***
	(0.0058)	(0.0015)
USMCA	-0.0058	-0.0165***
	(0.0038)	(0.0012)
EU	-0.0454***	0.0017*
	(0.0026)	(0.0009)
China	0.0037	0.0053***
	(0.0056)	(0.0018)
Observations	106,874	106,870
Adjusted R^2	0.00383	0.03424

Notes: Results of linear regressions of proportion of NTBs and average tariff exposure on trade blocs. Heteroskedastic-robust standard errors in parentheses. The "Other" bloc is the reference.

We include these origin bloc dummies in our main specification model, whose regression is shown in Table 10 (i.e., it examines whether the effects of NTBs and tariffs vary across major trade blocs). The coefficient on Tariff × NTB is negative and significant across all specifications, with values ranging from -1.56 to -2.51, confirming that NTBs reduce tariff responses regardless of trade bloc membership. However, the triple interaction term Tariff × NTB × MERCOSUR is positive, and it reaches statistical significance in more complete models. It suggests that trading within MERCOSUR may have greater flexibility in adjusting its reported trade values in response to NTBs and tariffs. In particular, we estimate an elasticity of customs duty evasion for transactions with NTBs and from Mercosur in values of 2,63 (3.15-2.51+0.12+1.87, column 9) and 2.1 in quantity (2.23- 2.4 + 0.26 + 2.01, column 11). A qualitatively similar but weaker effect is observed for China's unreported trade, where the interaction term is positive and marginally significant in some models that involve quantities.

Table 10: Regression results of interacting NTB and Trade Blocs.

	Baseline				Administrative				States			
Dep. Var.:	Value	Ext. Value	Quantity	Ext. Quantity	Value	Ext. Value	Quantity	Ext. Quantity	Value	Ext. Value	Quantity	Ext. Quantity
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Tariff	3.46***	2.44***	2.55***	2.37***	3.13***	2.17***	2.19***	1.94***	3.15***	2.19***	2.23***	2.01***
	(0.31)	(0.29)	(0.46)	(0.62)	(0.29)	(0.27)	(0.40)	(0.49)	(0.30)	(0.27)	(0.41)	(0.48)
Tariff X NTB	-2.20***	-1.56***	-2.04***	-1.80***	-2.44***	-1.71***	-2.33***	-2.12***	-2.51***	-1.74***	-2.40***	-2.15***
	(0.49)	(0.34)	(0.59)	(0.58)	(0.52)	(0.36)	(0.64)	(0.59)	(0.52)	(0.36)	(0.64)	(0.59)
Similar	-0.74***	-0.66***	-0.77**	-0.73**	-0.66***	-0.62***	-0.68**	-0.66**	-0.67***	-0.62***	-0.70**	-0.68**
	(0.23)	(0.16)	(0.31)	(0.29)	(0.23)	(0.16)	(0.30)	(0.28)	(0.23)	(0.16)	(0.30)	(0.28)
NTB	-0.11	-0.15*	-0.07	-0.11	0.08	-0.02	0.20	0.10	0.12	0.01	0.26	0.13
	(0.12)	(0.09)	(0.15)	(0.17)	(0.11)	(0.08)	(0.16)	(0.18)	(0.12)	(0.09)	(0.17)	(0.19)
Tariff X NTB X MERCOSUR	1.16	0.91	0.86	0.90	1.91*	1.45**	2.06**	2.27**	1.87*	1.43**	2.01**	2.20**
	(1.09)	(0.57)	(0.99)	(1.05)	(1.12)	(0.62)	(0.98)	(1.05)	(1.11)	(0.62)	(0.98)	(1.06)
Tariff X NTB X USMCA	1.07	0.76	0.17	0.80	1.01	0.74	0.18	0.91	1.04	0.72	0.16	0.86
	(1.04)	(0.95)	(1.17)	(1.24)	(1.08)	(0.98)	(1.24)	(1.29)	(1.07)	(0.96)	(1.23)	(1.27)
Tariff X NTB X EU	0.45	-0.01	0.07	-0.11	0.65	0.10	0.21	0.06	0.67	0.08	0.22	0.04
	(0.62)	(0.45)	(0.76)	(0.73)	(0.65)	(0.49)	(0.79)	(0.75)	(0.66)	(0.49)	(0.79)	(0.75)
Tariff X NTB X China	0.74	0.58	0.94	0.86	0.94	0.65	1.23	1.10*	0.95	0.66	1.30	1.16*
	(0.56)	(0.40)	(0.78)	(0.62)	(0.60)	(0.42)	(0.83)	(0.66)	(0.59)	(0.41)	(0.83)	(0.66)
Avg. Tariff	0.19	0.19	0.19	0.19	0.19	0.19	0.19	0.19	0.19	0.19	0.19	0.19
Avg. NTB	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27
Num.Obs.	59732	60908	54125	55784	59513	60687	53944	55590	59513	60687	53944	55590
R2 Adj.	0.114	0.180	0.180	0.306	0.119	0.185	0.185	0.311	0.120	0.187	0.188	0.313
FE: c	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
FE: p	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
FE: t	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Results of linear regressions interacting tariffs, NTB and trade blocs. The reference bloc is 'all other countries'. The administrative dataset is used. Two-way clustered (product and country) standard erros in parentheses.

Conversely, the interaction terms for USMCA and the European Union are consistently close to zero and statistically insignificant, indicating that firms trading with these regions do not exhibit differential unreported trade responses to NTBs in the presence of tariffs compared to the baseline group - Others. Overall, the findings suggest that while NTBs generally curb tariff evasion, their effects may vary across trade blocs, with firms importing from MERCOSUR and China showing some degree of adjustment in response to these trade policy measures. The robustness of these results across different specifications reinforces the role of NTBs as an effective enforcement.

Discussion. Brazilian trading with Mercosur partners is frequently characterized by longstanding commercial relationships and regional supply chains, often involving repeated transactions and higher levels of trust between parties. This may increase the opportunity for coordinated misreporting between exporters and/or importers (Fisman and Wei 2004). Moreover, imports from Mercosur countries often enter Brazil through land borders or designated customs corridors, which may be less intensively monitored than major seaports handling intercontinental trade, thereby reducing the perceived risk of detection. This would align with findings in the literature that show enforcement capacity varies by entry point, affecting evasion behavior (Javorcik and Narciso 2008).

5 Conclusion

This study highlights the significant impact of non-tariff barriers (NTBs) on unreported trade responses to customs duties in Brazil. We leverage a unique and confidential administrative dataset encompassing comprehensive transaction-level records of Brazilian imports and an exogenous tariff shock. We demonstrate that NTBs can modulate the elasticity of customs duties evasion (ECDE).

Our findings reveal that while higher tariffs increase unreported trade responses, the presence of NTBs can mitigate this effect. In addition, our data allows us to estimate that Brazilian importers resort to value- and quantity-based evasion. Therefore, the non-tariff barriers play a significant role in curbing the flexibility of customs duty evasion, emphasizing their unintended effectiveness in this context. These insights contribute to a deeper understanding of how NTBs influence trade dynamics and tax enforcement, offering valuable implications for policymakers aiming to enhance trade and tax compliance.

Our heterogeneous results provide further insights into the interaction between NTBs, tariffs, and firm- and product-level characteristics. First, manufacturing firms are susceptible to NTBs when combined with tariffs, highlighting the additional trade frictions imposed on these firms. Furthermore, in sectors where manufacturing firms dominate the restrictive effects of NTBs are exacerbated, further limiting unreported trade. This evidence highlights the differential impact of trade policies across industrial sectors, suggesting that manufacturing firms may not require additional regulatory barriers. Similarly, our results indicate that the impact of NTBs varies across product categories. The inclusion of product's economic categories (BEC) reveals essential distinctions. Specifically, using consumption goods as a baseline, intermediate, and capital goods face significant unreported trade reductions when tariffs and NTBs are present, and consumption-intermediate goods are less affected. These findings suggest that supply chain rigidity and regulatory constraints in the middle of the production chain play a crucial role in determining the extent to which NTBs hinder unreported trade. Additionally, our analysis of trade-bloc heterogeneity shows that firms importing from MERCOSUR and China display some unreported trade adjustments in response to NTBs and tariffs. In contrast, firms importing from

the USMCA or the European Union blocs exhibit similar unreported trade patterns compared to the benchmark. These results highlight the importance of considering trade policy heterogeneity across sectors, product categories, and trade partnerships when designing regulatory frameworks to balance trade facilitation with enforcement objectives.

NTBs reduce tariff underreporting by increasing compliance costs and regulatory oversight, making it more difficult and costly for firms to misreport trade values or quantities. Their deterrent effect is weaker for products intended for consumption, firms other than manufacturers, and goods from MERCOSUR, suggesting that institutional frameworks shape firms' ability to adjust reporting strategies in response to NTBs and tariffs. Our findings, therefore, indicate that NTBs can serve as effective instruments to curb customs duty evasion by limiting opportunities for misreporting. From a policy perspective, this suggests that governments might strategically design and implement NTBs not only to regulate trade flows but also to enhance revenue collection. Nevertheless, policymakers must remain cautious: while NTBs may deter evasion, they also risk creating additional frictions that distort legitimate trade and raise compliance costs. Any use of NTBs as an enforcement tool should thus be carefully balanced against potential efficiency losses in international trade. A promising avenue for future research is to examine how enforcement policies—particularly non-monetary factors such as auditor staffing, port infrastructure, or regulatory shifts—shape unreported gaps in import reporting and tariff evasion.

Supplementary Material

Supplementary material is available at the Oxford Economic Papers Journal online. These are the data and replication files and the online appendix.

Input Data.

All input data files should be placed in the input/ directory. Required files include:

Administrative Data.

These files are not present in the repository due to confidentiality constraints. Interested researchers who wish to access these datasets should submit a formal request to the Receita Federal do Brasil (Brazilian Federal Revenue Service) using the Lei de Acesso à Informação (Law on Access to Information). This legal mechanism allows researchers to request data from government agencies. For more information and to submit a request, visit the official Receita Federal website or the Brazilian government's access to information portal (https://www.gov.br/acessoainformacao/pt-br/assuntos/pedidos).

- `2021_tariff.csv`: Year 2021 tariff data (for comparison).
- `adicoes_por_ano_YYYY.csv`: Import records by year (YYYY = 2001–2006).
- `adquir_cnpj.csv`: Purchaser's CNPJ codes.
- `adquir_manufacture.csv`: Purchaser's manufacturing firm status.
- `horas_desemb.csv`: Customs clearance times per import.
- `importers_inic_ativ.csv`: Importers' initial activity classification.
- `unidades_rfb.csv`: RFB unit codes.

Note: The `adicoes_por_ano_YYYY.csv` files are the main administrative data source containing import declarations and are required to run the pipeline. The `unidades_rfb.csv` file contains customs unit information.

Trade Data.

- `TipoDI.csv`: Import declaration type codes.
- `trade_dt.fst`: Trade transaction data (FST format for efficient storage).
- `dt_y.csv`: Yearly trade data.
- COMTRADE data files:
  - `mishra_expo2001.csv` through `mishra_expo2006.csv`: Export data.
  - `mishra_impo2001.csv` through `mishra_impo2006.csv`: Import data.

Classification and Reference Data.

- `bec.csv`: Broad Economic Categories classification.
- `Rauch_classification_revised.csv`: Rauch product classification.
- `ncm_exemp.csv`: NCM exemption codes.

- `mfn_tariffs.csv`: Most Favored Nation tariff rates.

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Conflict of Interest

Rafael Bressan is an employee of the Brazilian Revenue Service (Receita Federal do Brasil - RFB). At no time was confidential data ever outside of the RFB's computing environment. The RFB has reviewed the manuscript to ensure it does not reveal confidential taxpayer data. The RFB has authorized the manuscript to be published, although the dataset and codes have to remain confidential. The purpose of this process is to ensure that researchers are not inadvertently disclosing confidential taxpayer information. Other than the above-stated disclosure, Rafael Bressan declares that he has no relevant or material financial interests that relate to the research described in this paper.The authors declare that they have no conflicts of interest.

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